vcolcanoplot

2026-03-13

```r
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.4.3
```

```r
library(ggrepel)
library(RColorBrewer)
library(grid)
library(scales)
```

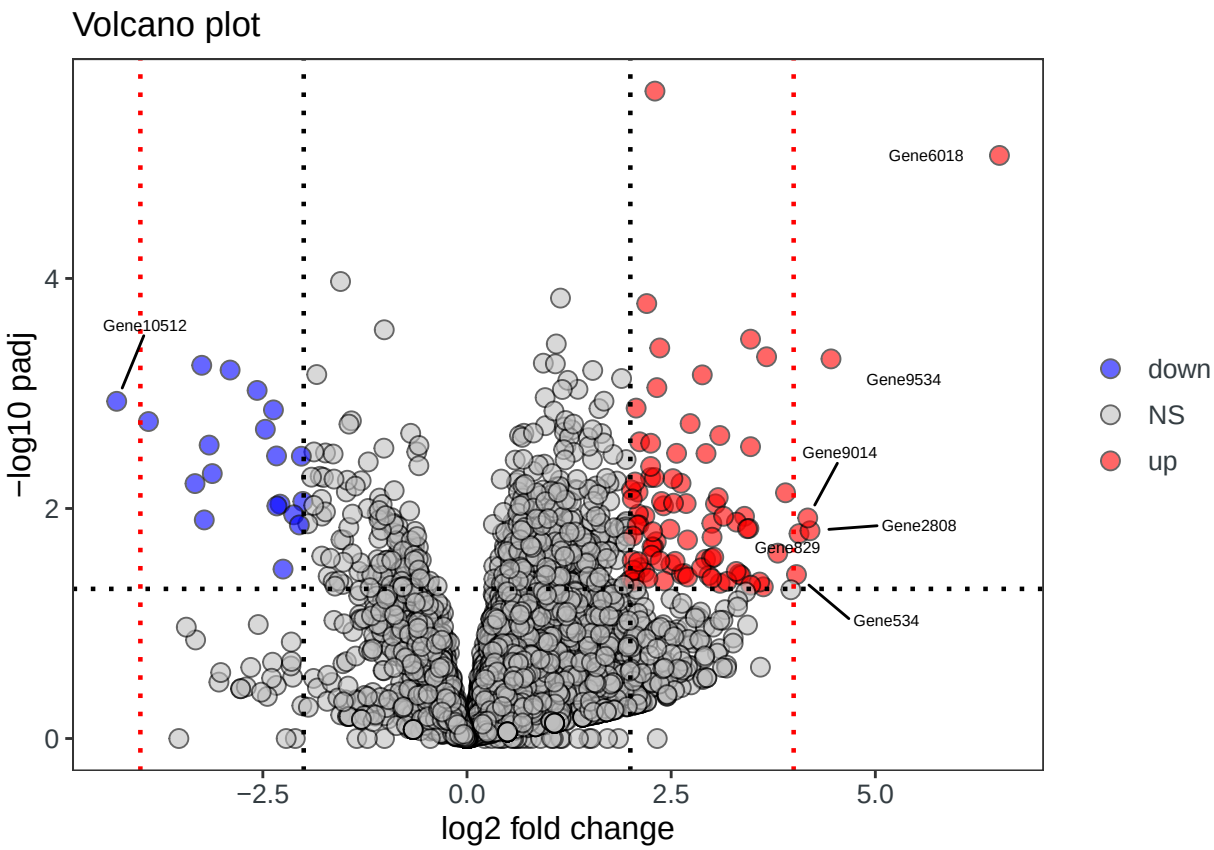
```r
df <- read.table("data.txt", sep="\t", header=TRUE, check.names=FALSE)

df$group <- as.factor(
  ifelse(df$padj < 0.05 & abs(df$log2FoldChange) >= 2,
         ifelse(df$log2FoldChange >= 2, "up", "down"),
         "NS")
)

df$label <- ifelse(
  df$padj < 0.05 & abs(df$log2FoldChange) >= 4,
  as.character(df$gene),
  ""
)
```

```r
p <- ggplot(df, aes(log2FoldChange, -log10(padj), fill=group)) +
  geom_point(color="black", alpha=0.6, size=3, shape=21) +
  theme_bw() +
  theme(panel.grid=element_blank(),
        axis.text=element_text(color='#333c41', size=10),
        legend.text=element_text(color='#333c41', size=10),
        legend.title=element_blank(),
        axis.title=element_text(size=12)) +
  geom_vline(xintercept=c(-2, 2), lty=3, color='black', lwd=0.8) +
  geom_vline(xintercept=c(-4, 4), lty=3, color='red', lwd=0.8) +
  geom_hline(yintercept=-log10(0.05), lty=3, color='black', lwd=0.8) +
  scale_fill_manual(values=c('blue', 'grey', 'red')) +
  labs(title="Volcano plot",
       x="log2 fold change",
       y="-log10 padj") +
  geom_text_repel(aes(label=label),
                  max.overlaps=10000,
                  size=2,
                  box.padding=unit(0.8,'lines'),
                  point.padding=unit(0.8,'lines'),
                  segment.color='black',
                  show.legend=FALSE)
```

```r
p
```




```r
ggsave(
  filename = "volcano_plot.png",
  plot = p,
  width = 6,
  height = 8,
  dpi = 300
)
```